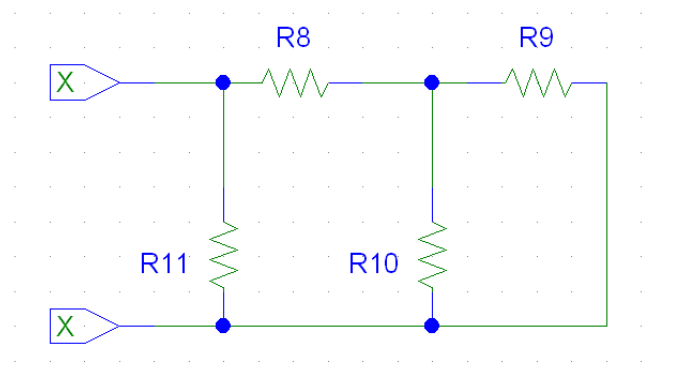


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Problem 01.html

Problem 01

Pr.1 Equivalent resistance

Due date: Wed Jan 26 11:59:00 pm 2022 (EST)



In the circuit, $R_8=45\text{ k}$, $R_9=40\text{ k}$, $R_{10}=30\text{ k}$, $R_{11}=35\text{ k}$;
The voltage source $V_s=16\text{ V}$ is connected to the terminals "X".

1) What is the equivalent resistance between the terminals "X"? (Unit: Ohm)

You are correct.

2) What is the total power delivered into the circuit by the voltage source V_s ? (Unit: W)

You are correct.

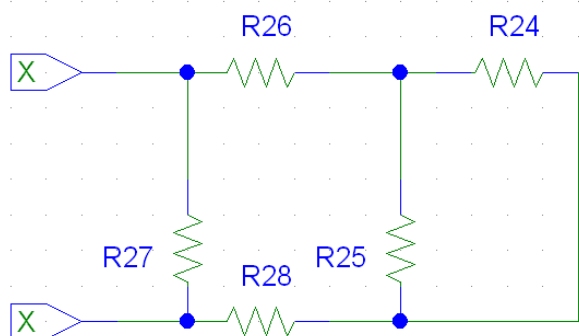
3) What is the power dissipated in the resistor R_{11} ? (Unit: W)

You are correct.

Problem 02.html

Problem 02

Pr.2 Equivalent resistance



In the circuit, each resistor is 97000 Ohm. Find the equivalent resistance between terminals "X" (unit: Ohm).

You are correct.

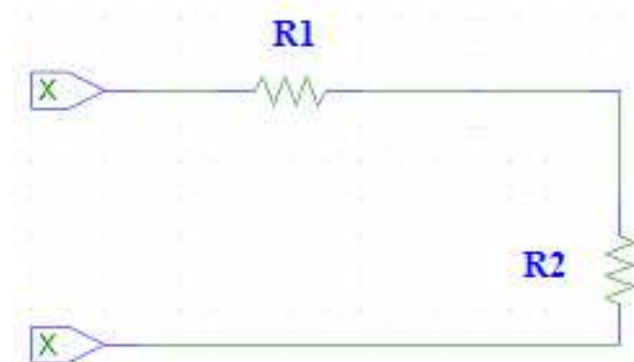
Problem 03.html

Problem 03

Pr.3 Voltage divider

The circuit below is used as a voltage divider. The input voltage is 12 V. The output voltage (that is the voltage across R_2) must be 5.1 V.

The current consumed by the divider must be 7.9 mA. Find the required resistances R_1 and R_2 . (Unit: Ohm)



$R_1 =$

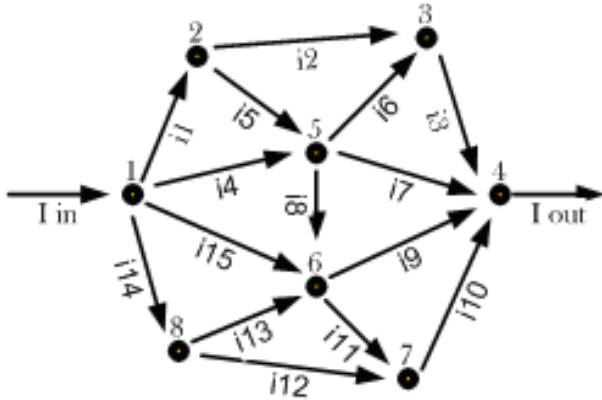
$R_2 =$

You are correct.

Problem 04.html

Problem 04

KCL



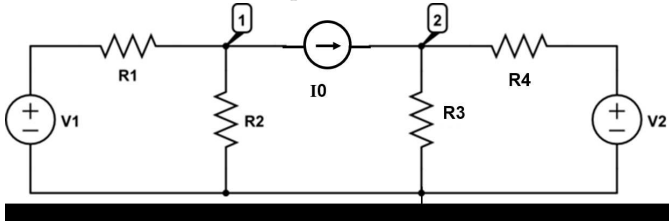
In the figure, if $I_{in} = 3.5$ A, $i_1 = 5.7$ A, $i_2 = 4.6$ A, $i_4 = 9.9$ A, $i_{15} = 5.2$ A, what is the current i_5 ? (UNIT:A)

You are correct.

Problem 05.html

Problem 05

Potenital via current 01.problem



In the circuit, $V_1 = 120$ V, $R_1 = 8$ Ohm, the current through R_1 flowing from left to right, $i_{R1} = 3$ A.

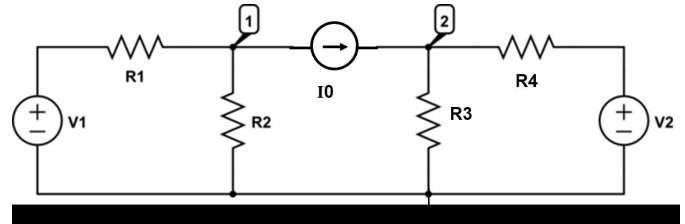
Find the potential of the node "1" (Unit: V)

You are correct.

Problem 06.html

Problem 06

Potenital via current 02.problem



In the circuit, $V_1 = 122$ V, $V_2 = 71$ V, $R_3 = 17$ Ohm, the current through R_3 flowing downwards, $i_{R3} = 2$ A.

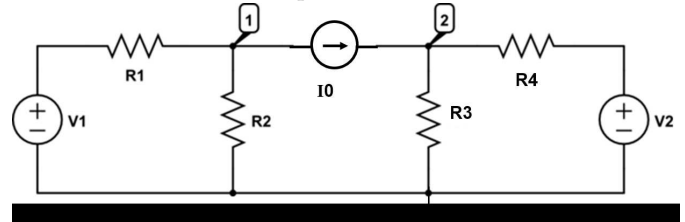
Find the potential of the node "2" (Unit: V)

You are correct.

Problem 07.html

Problem 07

Potenital via current 03.problem



In the circuit, $V_2 = 67$ V, $R_4 = 12$ Ohm, the current through R_4 flowing from left to right, $i_{R4} = 3$ A.

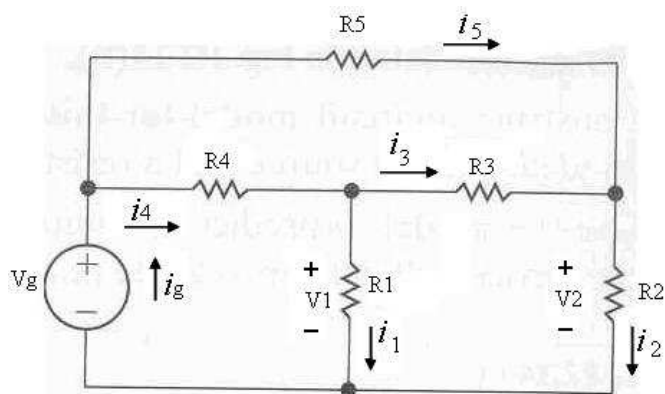
Find the potential of the node "2" (Unit: V)

You are correct.

Problem 08.html

Problem 08

KCL, KVL, Ohm's Law



In the circuit, $V_g = 108\text{ V}$, $R_1 = 11\text{ Ohm}$, $R_2 = 8\text{ Ohm}$, $R_3 = 9\text{ Ohm}$, $R_4 = 4\text{ Ohm}$ and $R_5 = 13\text{ Ohm}$.

The current $i_5 = 4.28\text{ A}$.

(A) What is the current i_1 ? (Unit: A)

You are correct.

(B) What is the power delivered by the voltage source V_g into the circuit? (Unit: W)

You are correct.

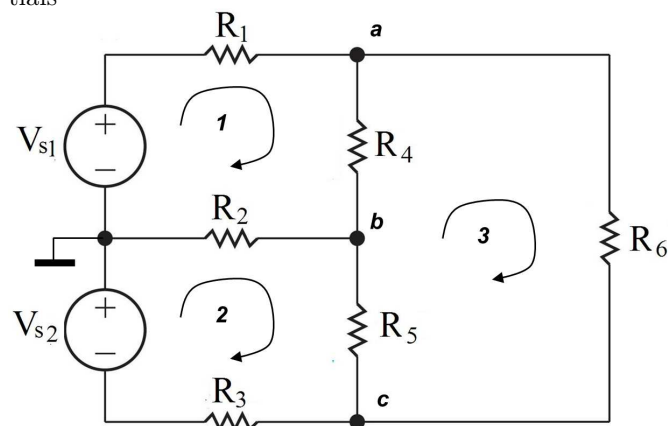
(C) What is the power dissipated in the resistor R_2 ? (Unit: W)

You are correct.

Problem 09.html

Problem 09

mesh equations formula response, mesh currents and potentials



In the circuit, $V_{s1} = 11\text{ V}$, $V_{s2} = 21\text{ V}$, $R_1 = 3.5\text{ Ohm}$, $R_2 = 7\text{ Ohm}$, $R_3 = 5.5\text{ Ohm}$, $R_4 = 5.5\text{ Ohm}$, $R_5 = 4\text{ Ohm}$, $R_6 = 5\text{ Ohm}$.

A. Assume that the mesh currents for the three meshes shown in the circuit diagram are defined as i_1 , i_2 and i_3 correspondingly.

Using the notations of the caption below circuit diagram (case sensitive!), write the mesh equations for the circuit keeping all the terms on the left-hand side (LHS), so that the equation appears as "LHS = 0"

Enter the left-hand side of mesh equation for the mesh "1"

= 0;

You are correct.

Enter the left-hand side of mesh equation for the mesh "2"

= 0;

You are correct.

Enter the left-hand side of mesh equation for the mesh "3"

= 0;

You are correct.

B. Using mesh analysis, answer the following questions.

1) Find the current through the resistor R_1 in the direction from left to right. (Unit: A)

You are correct.

2) Find the current through the resistor R_2 in the direction from right to left. (Unit: A)

You are correct.

3) Find the current through the resistor R_3 in the direction from right to left. (Unit: A)

You are correct.

4) Find the potential of the node "a" with respect to the ground node indicated in the circuit diagram. (Unit: V)

You are correct.

5) Find the potential of the node "b" with respect to the ground node indicated in the circuit diagram. (Unit: V)

Incorrect.

Tries 3/3

6) Find the potential of the node "c" with respect to the ground node indicated in the circuit diagram. (Unit: V)

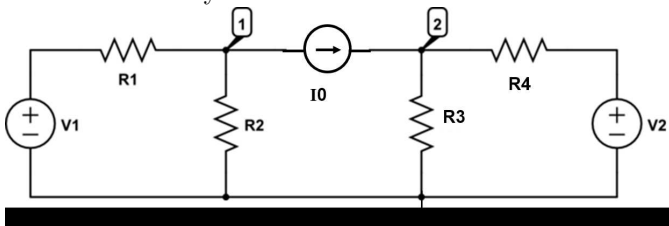
Incorrect.

Tries 3/3

Problem 10.html

Problem 10

Pr.10 Nodal analysis



In the circuit, $V_1=129$ V, $V_2=70$ V, $I_0=2$ A, $R_1=7$ Ohm, $R_2=50$ Ohm, $R_3=20$ Ohm, $R_4=23$ Ohm.

Solve the circuit using nodal analysis and answer the questions below.

1) Find the current through R_1 flowing in the direction from left to right. (Unit: A)

You are correct.

2) Find the current through R_2 flowing downwards. (Unit: A)

You are correct.

3) Find the current through R_3 flowing downwards. (Unit: A)

You are correct.

4) Find the current through R_4 flowing from left to right. (Unit: A)

You are correct.